18. FACIAL NERVE

- Is it a sensory, motor, or mixed nerve?
  - Mixed nerve (Sensory, motor & secretomotor root)
- What is the sensory root called?
  - Nerve of Wrisberg (Nervus intermedius)
- Where is the facial nerve nucleus present?
  - Pons
- What is the meaning of supranuclear and infranuclear palsy?
  - **Supranuclear** → Lesion in the cerebral cortex above the pons
  - **Infranuclear** → Lesion below the level of the nucleus in the pons
- Why is the forehead spared?
  - D/t B/L innervation from the cerebral cortex
  - **UMN palsy** → Forehead involved
  - **LMN palsy** → Forehead spared

ANATOMY OF FACIAL NERVE

00:02:01

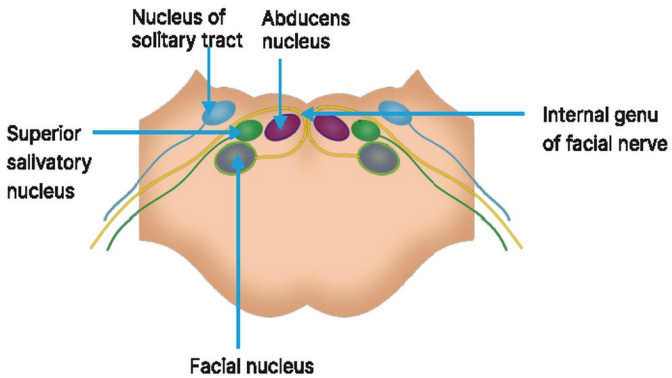
Subsegments of The Facial Nerve

INTRACRANIAL	• Nucleus in the pons → Internal auditory meatus (IAM)
INTRATEMPORAL	• IAM → Inner ear (labyrinthine segment) → Middle ear (tympanic segment) → Mastoid
EXTRACRANIAL	• Comes out of the mastoid through the stylomastoid foramen → Parotid gland → 5 terminal branches

Facial Nerve In The Brainstem

Facial Colliculus

- The facial nerve winds around the nucleus of the abducens, forming the facial colliculus & exits at the cerebello-pontine junction



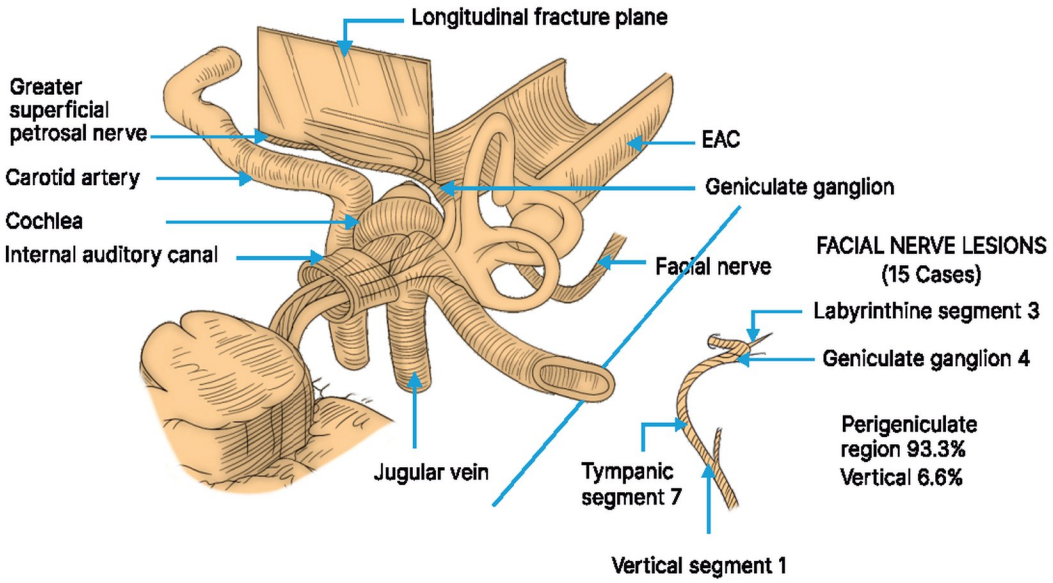
- **What happens when there is a lesion in the brainstem?**
  - 7th & 6th nerve palsy
  - C/L hemiplegia
    → This condition is also called Millard-Gubler syndrome

YOURWISH

YOURWISH

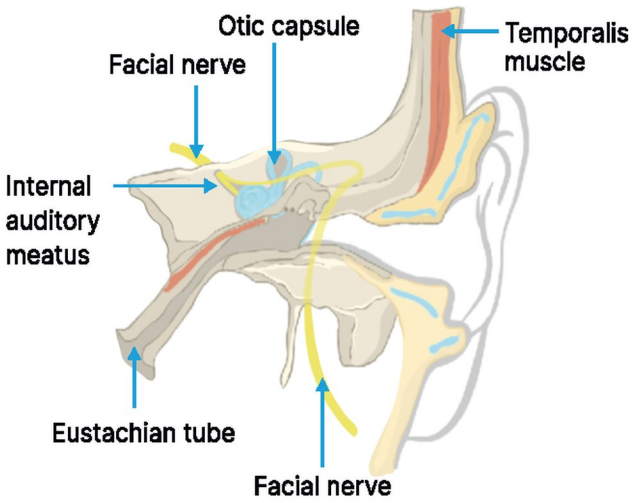
Intracranial Course Of The Facial Nerve

- Pons → Internal auditory meatus



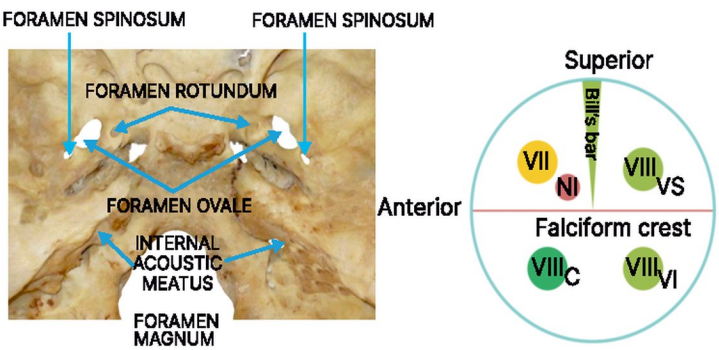
Intratemporal Segment of The Facial Nerve

- The intratemporal segment is the longest
- Only cranial nerve to run most of its course through a bony canal (Fallopian canal)
  - Bony canal → Dehiscent in 50%
  - Most common site of dehiscence → Horizontal part
- Divided into:
  - Meatal segment
  - Labyrinthine segment
  - Tympanic segment
  - Mastoid segment



Meatal Segment

- Length- 8-10 mm
- Seen in the IAM- Facial nerve travelling with the Cochlear nerve
- Vertical crest on bone in the superior part → Bill's bar
- Horizontal crest → Falciform crest
- Anterosuperior quadrant- Facial nerve
- Inferior quadrant- Cochlear nerve



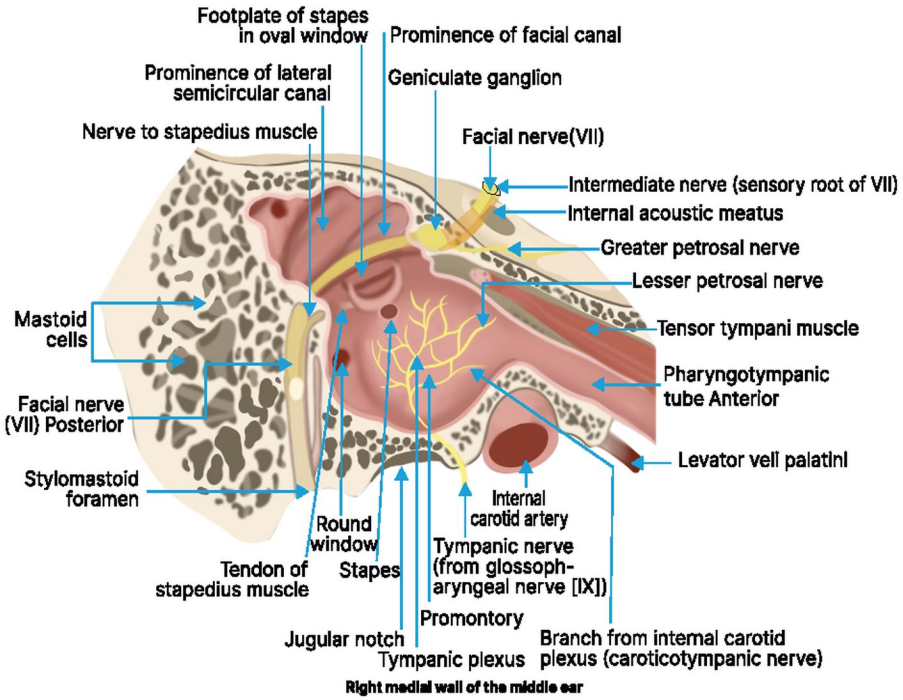
Labyrinthine Segment

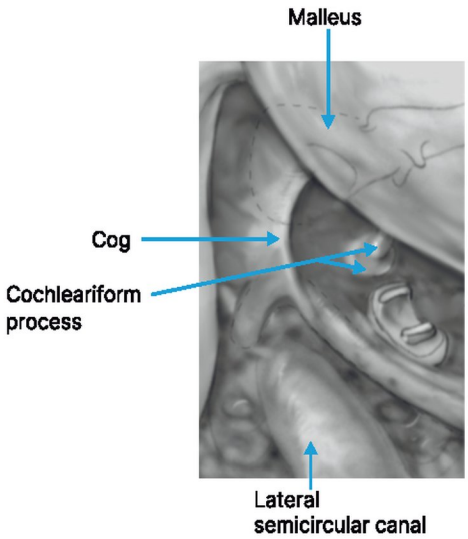
- Inner ear segment
- 3-4 mm in length and 0.7 mm in diameter → shortest & narrowest → prone to ischemia & edema (Bell's palsy)
- Travels over the vestibule in the inner ear
  - Anteriorly → Cochlea
  - Posteriorly → Semicircular canals

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Tympanic Segment

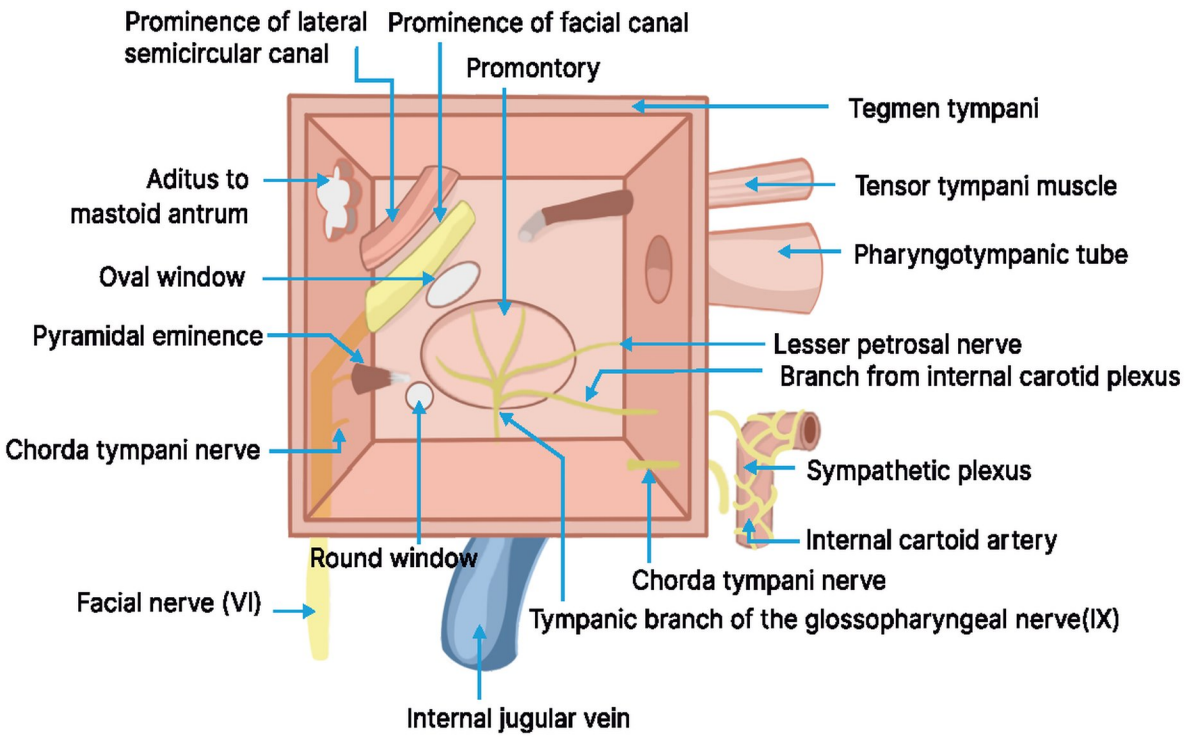
- Horizontal segment - 11 mm & Vertical - 13mm
- From the inner ear, it enters the medial wall of the middle ear, making a turn → First genu/Geniculate ganglion
  - Landmark for first genu: Cochleariform process
- Horizontal segment: Runs above the oval window & below the lateral SCC
- Goes to the mastoid to make the second turn: Second genu
  - Landmark: Pyramidal process





Mastoid Segment

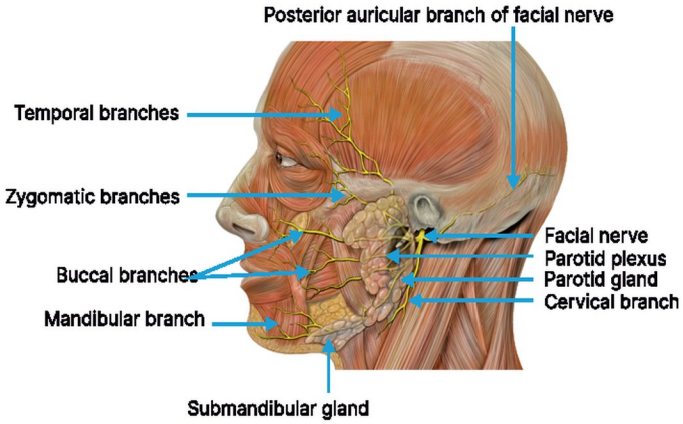
- The nerve runs vertically, also known as the mastoid segment
  - Landmark: Tympanomastoid suture



Extracranial Segment Of The Facial Nerve

00:10:07

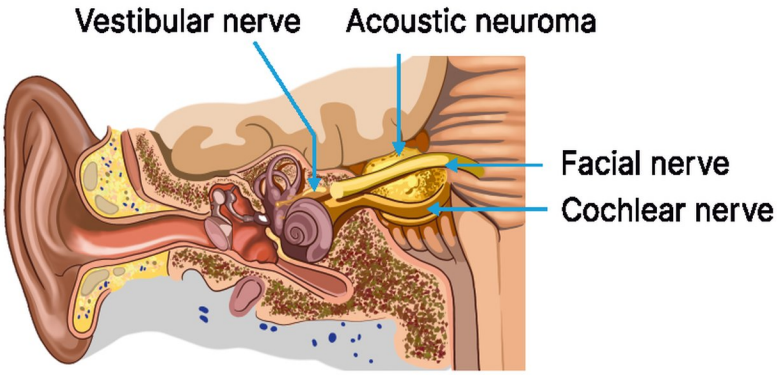
- Exits through the stylomastoid foramen → Between the superficial & deep lobes of the parotid gland → gives 5 terminal branches
  - Temporal, Zygomatic, Buccal, Marginal mandibular & Cervical



Subsegments & Features

Part	Special Feature	Landmarks	Branches
Meatal segment	Present within the Internal Auditory Meatus (IAM).	Separated from the superior vestibular nerve by a bony ridge called Bill's bar	None
Labyrinthine segment	Located in the inner ear, the shortest and narrowest segment of the facial nerve canal		None
Tympanic / Horizontal segment	Runs along the medial wall of the middle ear; a common site for congenital dehiscence of the fallopian canal	Processus cochleariformis Cog Between the bulge of lateral SCC above & oval window below	1st **branch - Greater superficial petrosal nerve**: arises from geniculate ganglion; controls lacrimation and nasal & palatal secretions via sphenopalatine ganglion; assessed by Schirmer's test.
Mastoid / Vertical segment	Located on the posterior wall of the middle ear, prone to iatrogenic injury during mastoid surgery	Nerve lies laterally and posterior to the pyramid	2nd **branch - Nerve to stapedius (Mediates stapedial reflex)**: lesion causes hyperacusis 3rd **branch - Chorda tympani**: taste from the anterior 2/3rd of the tongue and parasympathetic supply to the submandibular & sublingual glands; tested by electrogustometry & salivary flow 4th **branch - Sensory auricular branch**: supplies the pinna & external auditory canal; involved in the Hitzelberger sign (Acoustic neuroma)

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BRANCHES OF THE FACIAL NERVE

00:12:46

- GSPN → Lacrimation (Assessed by Schirmer's test)
- Nerve to stapedius → Stapedial reflex
- Chorda tympani → Taste from anterior 2/3rd
- Sensory auricular branch to pinna & EAC
- Branches to the submandibular & sublingual glands
- 5 terminal branches → Motor muscles of facial expression

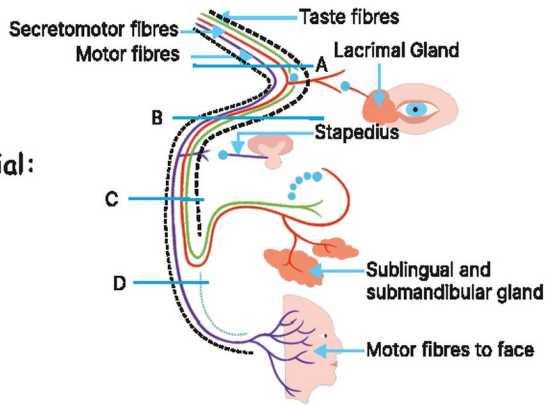
Topodiagnosis

A) Lesion above the first genu - Supragenicular lesion:
- Loss of lacrimation, stapedial reflex, taste on anterior 2/3rd, dry mouth & LMN type of facial palsy

B) Lesion at the horizontal segment/Infragenicular/Suprastapedial:
- Lacrimation is preserved & rest of the functions are lost

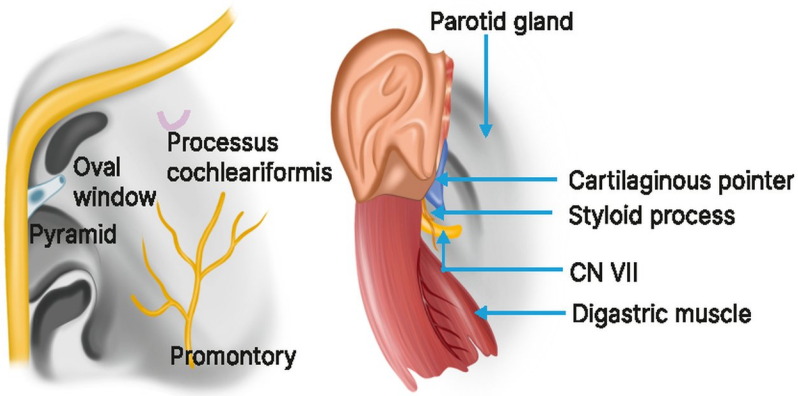
C) Lesion at the vertical/Mastoid segment:
- Schirmer's & acoustic reflex are preserved & the rest are lost

D) Lesion outside the temporal bone:
- Motor paralysis of the face, along with sparing of all the nerves



Surgical Landmarks

00:16:38



- Processus cochleariformis: The geniculate ganglion lies in front of it
- Oval window and Horizontal canal: The facial nerve lies between them
- Pyramid and fossa incudis
- Digastric ridge

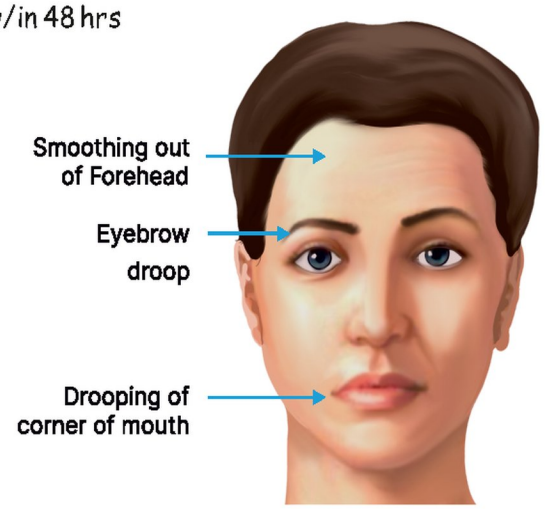
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Bell's Palsy

00:16:59

- Idiopathic facial nerve palsy
- Peripheral facial lesions are more common than central, and about **2/3**rd of peripheral lesions are **Bell's palsy**
- Risk is **3x greater** in pregnancy, especially in the **3**rd **trimester**
- Most commonly associated with **HSV** or **EBV**
- Increased risk with diabetes
- **Clinical features:**
  - Sudden onset symptoms, usually hours with maximal weakness w/in 48 hrs
  - Unilateral
  - Eyebrow sagging
  - Inability to close the eye
  - Loss of nasolabial fold
  - Decreased tearing
  - Hyperacusis
  - Loss of taste to the anterior 2/3rd of the tongue
  - Mouth droop
- **Diagnosis:** Is by exclusion
- **Supportive treatment:** Care of the eye and physiotherapy
- **Medical Management:** Prednisolone (1 mg/kg/day)
  - 10 days for recovery
  - 20 days if recovery is incomplete
- **Surgical Management:** Vertical, tympanic +/- labrynthine segments are decompressed
- **Prognosis: 90%** recover. Good recovery in incomplete Bell's palsy



Merkelson Rosenthal Syndrome

- Recurrent facial nerve palsy
- Swelling of the lips
- Fissured tongue


Swelling of lips


Fissured tongue

FMGE Jan 2023

Recurrent Facial Nerve Palsy

- Melkerson syndrome
- Diabetes
- Sarcoidosis
- Tumours

Ramsay Hunt Syndrome/herpes Zoster Oticus

- **Treatment:** Acyclovir/Valacyclovir + Steroids
- Poorer prognosis than Bell's palsy


Vesicles over External

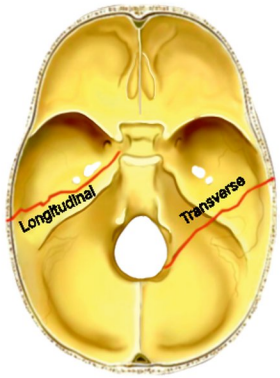

Facial nerve palsy

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FRACTURES OF THE TEMPORAL BONE

00:19.38

- Which is common?
  - Longitudinal
- TM perforation occurs commonly in?
  - Longitudinal
- Facial nerve palsy is higher in?
  - Transverse (Facial nerve runs parallel to the longitudinal)
- CHL & CSF otorrhea seen in → Longitudinal
- SNHL seen in → Transverse
- CSF rhinorrhea seen in → Both Longitudinal & Transverse



ELECTROPHYSIOLOGICAL TESTS

00:21:30

Electroneuronography	Electromyography
• If >90 decrease in CMAP within 14 days of paralysis, then the chances of spontaneous recovery is negligible - consider surgical decompression	• Procedure Interpretation - if motor unit potentials are present, then it means the nerve is regenerating • If fibrillation potentials are present, then the chances of spontaneous recovery is negligible • No electric output, then facial reanimation may be considered

Tests for Facial Nerve Palsy

00:22:35

- Topodiagnostic tests - Used to determine the site of nerve lesion:
  - Schirmer test → Assesses lacrimation (dry eye)
  - Stapedial reflex → Absence suggests hyperacusis
  - Evaluation of taste and salivation
- Electrophysiological tests - Helpful in assessing prognosis:
  - Electroneuronography (ENoG): Should not be performed before 3 days, as Wallerian degeneration begins after this period.
  - Electromyography (EMG): Usually performed after >3 weeks

ONE LINERS

- First branch of the facial nerve → GSPN
- Nerve involved in hyperacusis → Nerve to stapedius
- Lacrimation is affected when the facial nerve is injured at what level → 1st genu
- Taste is affected when the facial nerve is injured at what level → Vertical
- The most common cause of facial nerve palsy → Idiopathic / Trauma
- Iatrogenic facial palsy occurs most commonly in which surgery → Mastoid
- The segment of the facial nerve affected first in ischemia and edema → Labyrinthine segment
- Which segment of the facial nerve is affected first in iatrogenic trauma → 2nd genu/Mastoid
- Which fracture of the petrous bone is responsible for facial nerve injury → Transverse
- Crocodile tears occur due to faulty reinnervation with → Gustatory lacrimation due to faulty innervation of GSPN with the chorda tympani nerve

IMPORTANT PYQs ON TRAUMATIC FACIAL NERVE PALSY

- MC iatrogenic cause - Parotid Sx
- 2nd MC iatrogenic cause - Mastoid Sx
- MC site injured in mastoid Sx → 2nd genu followed by vertical segment
- Non-iatrogenic Trauma - More common in a transverse fracture of the temporal bone
- Overall incidence of Longitudinal > transverse fracture
- Ecchymosis behind the ear following basilar skull fracture - k/a Battle's sign
- Best investigation for petrous temporal bone fracture - HRCT
- Following trauma, late-onset palsy management - Steroids
- Following trauma, sudden onset palsy management - Re-exploration and repair
- Aberrant regeneration of the facial nerve can lead to crocodile tears and synkinesis.

SYNDROMES ASSOCIATED WITH THE FACIAL NERVE

- Millard-Gubler
- Melkersson-Rosenthal
- Heerfordt syndrome
- Ramsay Hunt syndrome

FAULTY INNERVATION

- **Crocodile tears** - Injury is before the geniculate ganglion, and reinnervating fibres anastomose with the CT nerve
- **Frey's syndrome** - Gustatory sweating - faulty reinnervation with AT nerve

PYQs

00:25:15

Q. Which diagnosis is most likely for a patient experiencing painful vesicles in the external auditory meatus and over the tympanic membrane, along with facial nerve palsy? Furthermore, which specific site is affected in this case?

a. Ramsay Hunt syndrome and basal ganglion FMGE Jan 2023
b. Ramsay Hunt syndrome and geniculate ganglion
c. Melkersson syndrome and the trigeminal ganglion
d. Melkersson syndrome and otic ganglion

Ans: **b.** Ramsay Hunt syndrome and geniculate ganglion

- Virus remains latent in the geniculate ganglion

YOURWISH

19. INSTRUMENTS

INSTRUMENTS USED IN TONSILLECTOMY

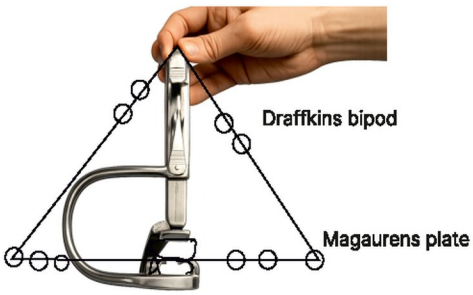
00:00.19

Boyle Davis Mouth Gag With Tongue Blade

- Self-retracting mouth gag
- Stabilised by Draffkin's bipod on Magaurens plate
- The upper end with two blades retracts the upper jaw
- The lower end with the tongue blade retracts the lower jaw







Mollison Tonsillor Dissector

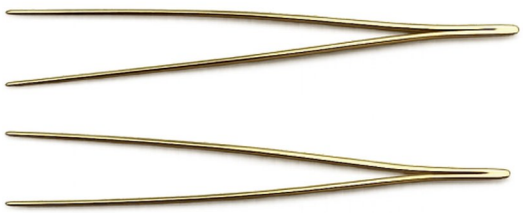
- The tonsil is pulled medially with tonsil holding forceps and dissected from the tonsillar bed using Mollison tonsillor dissector
- This has also got an anterior pillar retractor to retract the anterior pillar and visualise the bed of tonsil to check for a bleeder



Mollison pillar retractor with dissector

Waughs Tenaculum Forceps

- To hold the tissue
- Can be toothed or untoothed



Waugh's Tenaculum forceps

Eve's Tonsillar Snare

- Used to dissect the lower pole of the tonsil
- The pedicle is engaged in the snare and pulled so that the base of the tonsil is crushed and cut



Yonkos Suction

- Used to suction the bed of the tonsil to clear the bleeding



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INSTRUMENTS USED IN ADENOIDECTOMY

00:02.40

St Clair Thompson Adenoidectomy Curette



- It has a detachable cage
- Held like a dagger, goes from the oropharynx to the nasopharynx above, and the adenoid is churned off from bed
- Initially used with a cage to prevent chunks of adenoid aspiration
- Later used without a cage to smoothen the bed of tonsil

BULLS LAMP AND HEAD MIRROR



- The head mirror is concave
- Reflects the light from the bull lamp onto the area of interest

FREERS ELEVATOR AND KELLIANS ELEVATOR



- To palpate or elevate the tissue

JOBSONS HORNE PROBE WITH RING CURETTE



- To remove wax or a foreign body

SIEGELS PNUEMATIC SPECULUM



- To assess the mobility of the tympanic membrane
- To increase the pressure in the EAC to elicit the fistula sign
- The speculum end goes into the EAC
- Cuffed end to increase and decrease the pressure

OTOSCOPE

- To visualise the outer ear and tympanic membrane
- Magnification of 2 to 2.5 times is obtained



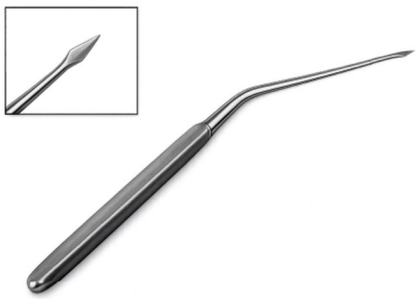
SIMPSONS AURAL SYRINGE

- Used in syringing to remove the non-hygroscopic foreign body
- It is directed posterosuperiorly to avoid hitting the Tympanic membrane directly
- The temperature of the water used is of body temperature
- 7°C above or below the body temperature can be used, but elicits a caloric reaction in the caloric test



MYRINGOTOME

- Used to make hole in the tympanic membrane to drain the pus in ASOM OR CSOM



GROMMET

- Ventilating tube used in serous otitis media



MASTOID RETRACTORS

Mollison Self-Retaining Hemostatic Mastoid Retractor

- An equal number of prongs on both blades
- Eg: 4 and 4 or 3 and 3



Withlaner Self-Retaining Hemostatic Mastoid Retractor

- An unequal number of prongs on both blades
- Eg: 4 and 3 or 3 and 2



Flesters Self-Retaining Hemostatic Mastoid Retractor

- 4 prongs on one side and 3 on the other side, but one prong is longer than the other two



Perkins Self-Retaining Hemostatic Mastoid Retractor

- 1 prong on one side and 3 OR 4 on the other side



INSTRUMENTS USED IN NASAL EXAMINATION

Thudicums Nasal Speculum



Killians Nasal Speculum

- Self-retaining speculum used in surgery
- Can be used for examination, diagnostic, and therapeutic purposes



Ring Currette

- Used to remove foreign bodies or any other pathologies



Ballinger's Sweivels Knife

- Was used in SMR, which is not done nowadays
- Not used in septoplasty
- Cuts the cartilage and bone with its rotatory motion



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Endoscopes

Rigid Endoscopes

- Available in 0°,45°,70°,90°



Fibreoptic Flexible Endoscopes

- To visualize the vocal cords' mobility and their vibration



Asher's Forceps Vs Walshams Forcep

ASHER'S FORCEPS	WALSHAM'S FORCEP
• Asher's forceps is angulated • Used in nasal septal reduction • Both prongs are bare	• Walsham's is straight • Used in nasal bone reduction • It has got rubber tubing on one of the prongs to avoid injury to the orbit • The prong with rubber tubing lies outside the nasal cavity, while the bare one is inside the cavity





Lichwitz Trochar And Canula

- Used for antral wash when the maxilla is opened
- Not used now



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